MITHRAKAYAL M

mithrakayal137@gmail.com | 7598371986

LinkedIn: linkedin.com/in/mithrakayal-mani-b56b7a377 | GitHub: https://github.com/MITHRAKAYAL137

PYTHON DEVELOPER

Aspiring Data Scientist | Machine Learning Engineer

PROFESSIONAL SUMMARY

• **Python Developer** with **2+ years of experience** building **real-time data processing systems, automation tools, and computer vision applications**. Skilled in **Python, OpenCV, NumPy, SQL, and API integration** with hands-on experience in **dashboard validation, data extraction, and system optimization** for **EV and healthcare domains**. Proven ability to deliver **efficient, reliable, and high-accuracy solutions** in complex projects.

PROFESSIONAL EXPERIENCE

Python Developer – *Infiniti Automation, Coimbatore*

June 2023 – Present

EV Dashboard Validation & Data Extraction System

Version 1 – ML-Based Region Detection

- Developed a **machine learning-based computer vision system** to detect critical regions on EV dashboards
- Built **automated pipelines** for data extraction and validation under **varying conditions**
- Enabled **real-time monitoring** by sending processed data to backend systems

Version 2 – Numerical Processing & Image Verification

- Implemented **seven-segment digit recognition** using **NumPy-based numerical processing**
- Extracted **accurate numerical values** directly from dashboard images
- Integrated system to send **images and extracted data** to server for validation
- Improved **system reliability** through **dual validation** (visual + numerical)

Medical Device Monitoring & UI System (Confidential)

- Developed a **lightweight, high-performance GUI** with advanced **real-time data handling and multimedia integration**
- Enabled **dynamic monitoring and recording functionality** for smooth device operations
- Integrated **network connectivity features** for efficient backend communication and data transfer
- Optimized **system performance and reliability** through clean design and efficient backend integration

PROJECTS

AI Smart Traffic Monitoring System – Python, OpenCV, YOLOv8, Streamlit

- Built a computer vision system to detect and track vehicles in video streams for real-time traffic analysis.
- Developed an interactive Streamlit dashboard showing total vehicles, live traffic, and congestion levels.
- Logged vehicle and traffic data for actionable insights and future forecasting.

Urban Traffic Intelligence System – *Python, Streamlit, Plotly, Folium, ARIMA*

- Developed a real-time smart city traffic monitoring platform with interactive dashboards, heatmaps, and road network visualizations.
- Implemented AI-based traffic prediction and 24-hour forecasting using time series models, improving prediction accuracy.
- Simulated IoT traffic sensors and visualized congestion levels on network and city maps for data-driven traffic management.

AI-Driven Solar Dryer Optimization System – Python, ML, Data Analysis

- Developed predictive models to optimize solar dryer conditions for maximum efficiency.
- Applied feature engineering and statistical analysis to improve model accuracy.
- Enabled data-driven optimization for reduced drying time and energy consumption.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Scikit-learn, Supervised & Unsupervised Learning, Time Series Forecasting, ARIMA
Computer Vision: OpenCV, Image Processing, Region Detection
Data Tools: Pandas, NumPy, Data Cleaning, Feature Engineering
Visualization: Matplotlib, Seaborn, Streamlit
Systems & Concepts: Real-Time Data Processing, Data Validation, API Integration, Backend Systems
Tools: Git, Visual Studio Code (VS Code), Jupyter Notebook, Streamlit

KEY ACHIEVEMENTS

✓ **Reduced system downtime by 25%** by implementing **predictive monitoring** for EV dashboards.

✓ **Automated validation pipelines**, minimizing manual errors by **80%** in real-time data processing and device monitoring systems.

✓ **Developed a lightweight liquid glass UI** for medical device monitoring, achieving **high-performance, visually appealing dashboards** comparable to heavier frameworks.

✓ **Built real-time data extraction and processing pipelines** using Python and OpenCV for EV dashboards, improving **data reliability and system efficiency**.

EDUCATION

B.E – Computer Science and Engineering
Dhanalakshmi Srinivasan College of Engineering, **Coimbatore**
CGPA: 8.4 | 2019 – 2023

CERTIFICATIONS

- Python – QSpiders
- SQL – QSpiders